

Activity 8 - Use an array-backed ring buffer

Version2.0 /K3/A3 /C-compatible core, Arduino C++ adapter

Design contract

Design C-compatible logic for this exact contract: Buffer capacity is 4; count 0..4; next 0..3; oldest sample overwritten. Empty mean returns 0. Samples are restricted to -1000..1000. Keep Arduino APIs in the C++ adapter.

Vibe-code implementation

Implement Use an array-backed ring buffer. Preserve logic.h, units and thresholds. Explain every assumption. Add independently predicted boundary tests; do not invent hardware/API evidence.

Repair from evidence

Given the first failed assertion, explain expected versus actual, identify the smallest contract-preserving fix and add a regression test. Do not edit expected values to hide the defect.

Safe workflow

Use mock data only. Read the exact specification, request a small reviewable patch, check units and signatures, predict independent tests, compile and run, then record the human decision. Never turn AI text into shell, pump or arbitrary actuator commands.

Host tests do not prove physical upload, LCD/SD behaviour or local-model performance. Record board verification as NOT RUN until measured. Detailed procedures: Activity README and Learner Guide.